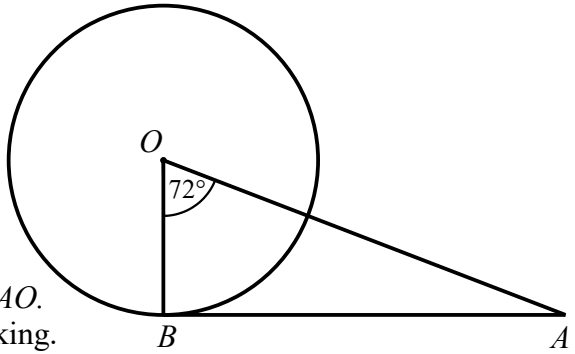
1 *B* is a point on the circumference of a circle, centre *O*.

AB is a tangent to the circle.

Angle *BOA* = 72°

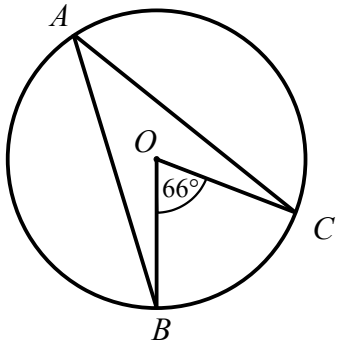


Work out the size of angle *BAO*.
You must show all your working.

(2 marks)

2 *A, B, C* and *D* are points on the circumference of a circle.

Angle *BOC* = 66°

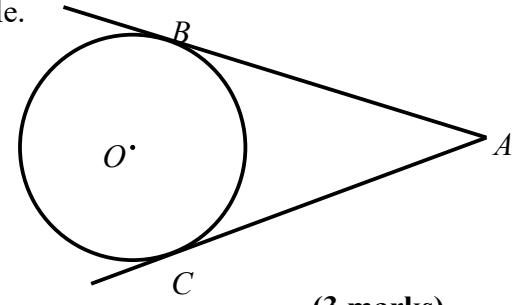


Find the size of angle *BAC*.
Give a reason for your answer.

(2 marks)

3 *B* and *C* are points on a circle, centre *O*.
AB and *AC* are tangents to the circle.

Angle *BAC* = 40°

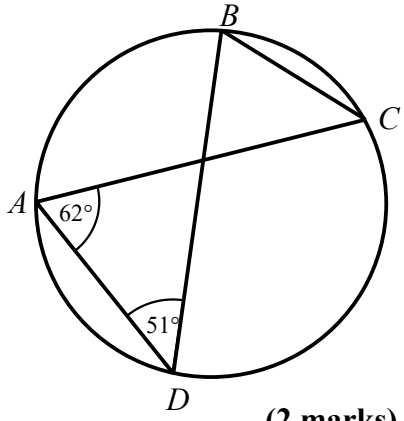


Work out the size of angle *BOC*.
You must show all your working.

(3 marks)

4 *A, B, C* and *D* are points on the circumference of a circle.

Angle *CAD* = 62°
Angle *ADB* = 51°

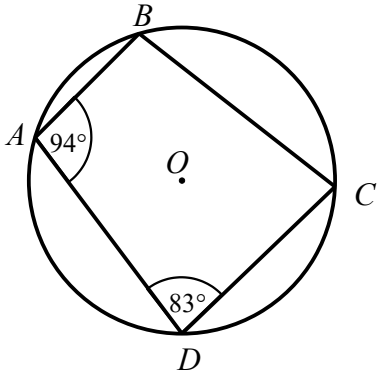


Find the size of angle *ACB*.
Give a reason for your answer.

(2 marks)

5 *A, B, C* and *D* are points on the circumference of a circle.

Angle *BAD* = 94°
Angle *ADC* = 83°

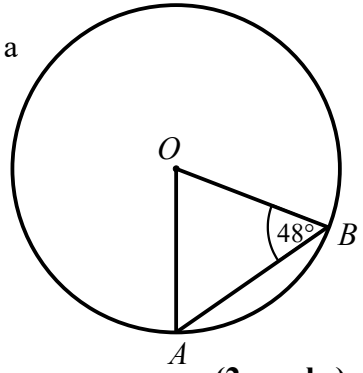


Find the size of angle *ABC*.
Give a reason for your answer.

(2 marks)

6 *A* and *B* are points on the circumference of a circle, centre *O*.

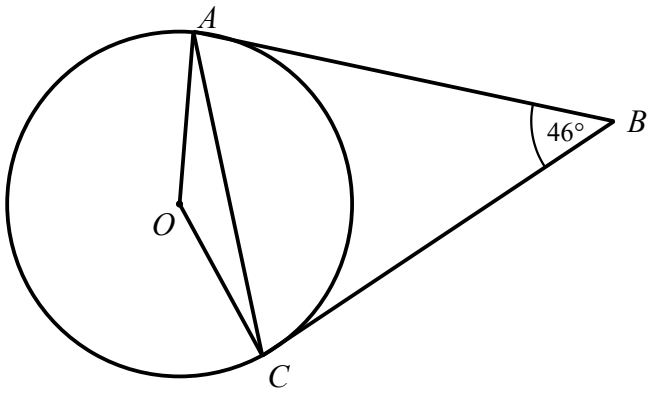
Angle *ABO* = 48°



Find the size of angle *AOB*.
Give a reason for your answer.

(2 marks)

7



A and *C* are points on the circumference of a circle, centre *O*.
AB and *BC* are tangents to the circle.

Angle *ABC* = 46°

Find the size of angle *OAC*.
Give reasons for each stage of your working.

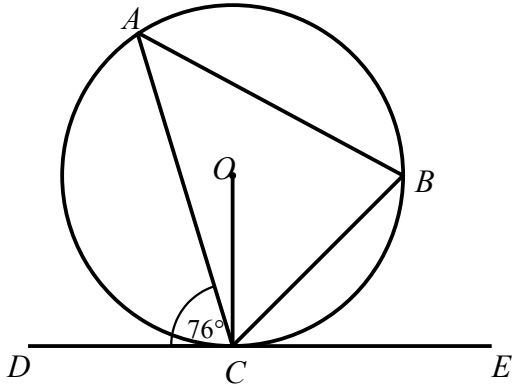
(4 marks)

8 *A* and *B* are points on the circumference of a circle, centre *O*.
DCE is a tangent to the circle.

Angle *ACD* = 76°

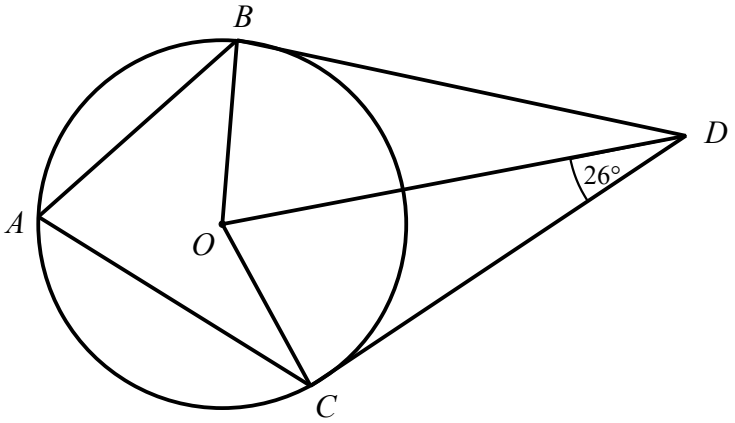
(a) Find the size of angle *ACO*.
Give reasons for each stage of your working.

(b) Find the size of angle *ABC*.
Give reasons for each stage of your working.



(4 marks)

9



A, B and *C* are points on the circumference of a circle, centre *O*.
BD and *CD* are tangents to the circle.

Angle *ODC* = 26°

Find the size of angle *BAC*.
Give reasons for each stage of your working.

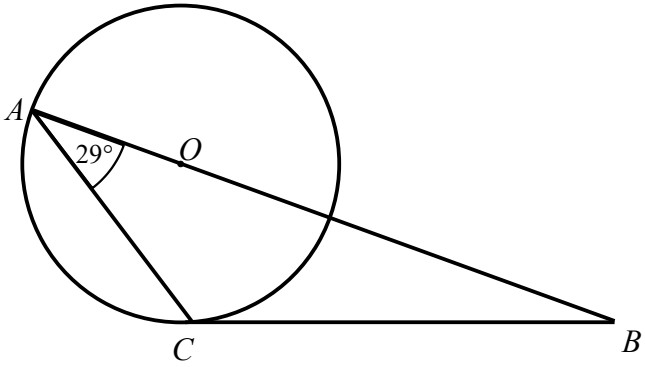
(4 marks)

10

A and *C* are points on the circumference of a circle, centre *O*.

BC is a tangent to the circle.

Angle *CAB* = 29°



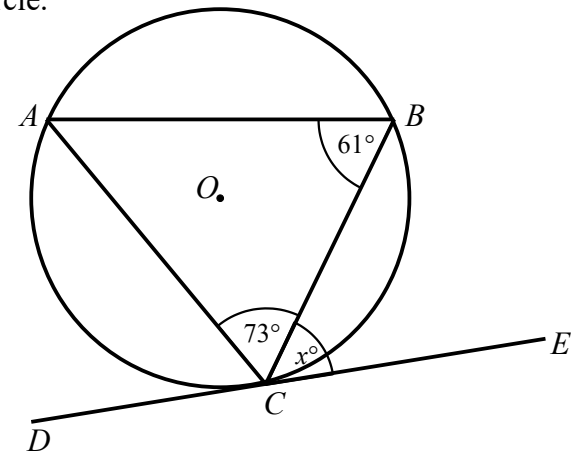
Find the size of angle *ABC*.
You must show all your working.

(4 marks)

11

A, B and *C* are points on the circumference of a circle, centre *O*.
DCE is a tangent to the circle.

Angle *ABC* = 61°
Angle *ACB* = 73°
Angle *BCE* = *x*°



Find the value of *x*.
Give reasons for each stage of your working.

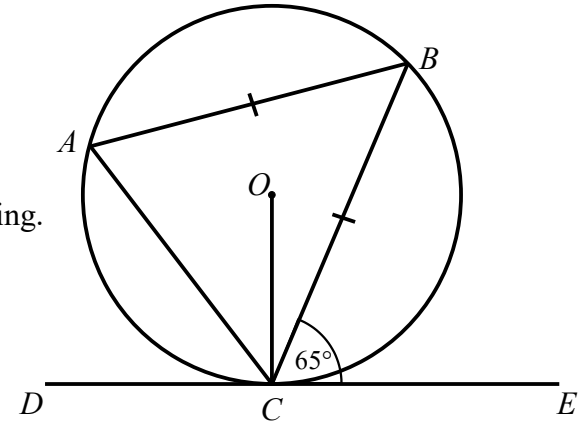
(3 marks)

12

A, B and *C* are points on the circumference of a circle, centre *O*.
DCE is a tangent to the circle.

AB = *BC*
Angle *BCE* = 65°

Find the size of angle *AOC*.
You must show all your working.



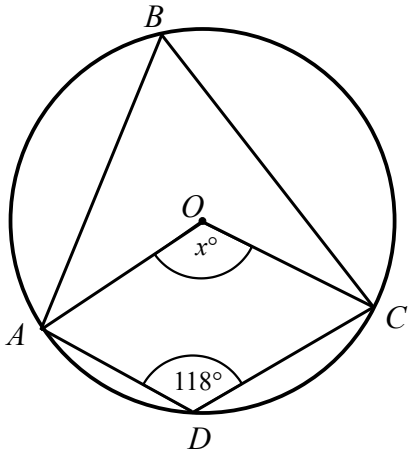
(4 marks)

13

A, B, C and *D* are points on the circumference of a circle, centre *O*.

Angle *ADC* = 118°
Angle *AOC* = *x*°

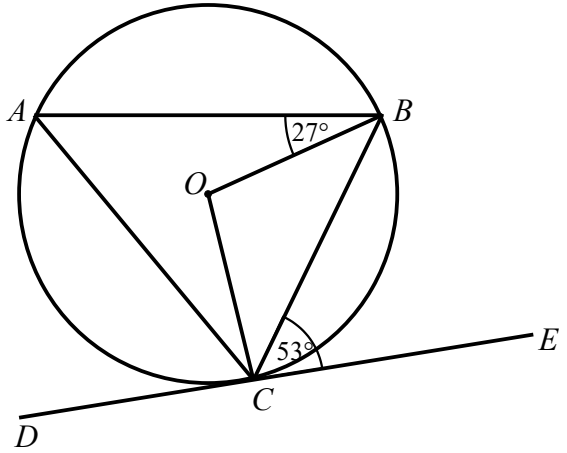
Work out the value of *x*.
You must show all your working.



(3 marks)

14 *A, B* and *C* are points on the circumference of a circle, centre *O*.
DCE is a tangent to the circle.

Angle *ABO* = 27°
Angle *BCE* = 53°



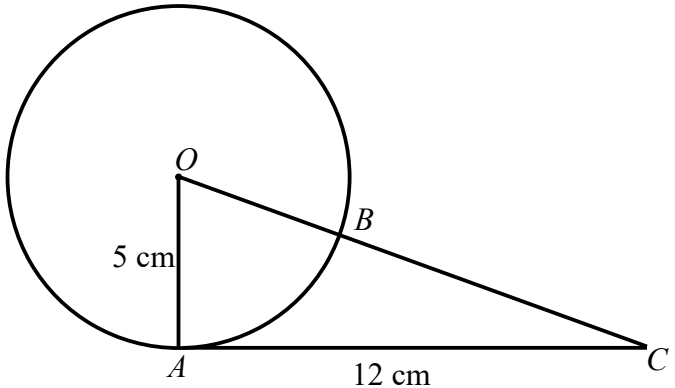
Find the size of angle *ACO*.
Give reasons for each stage of your working.

(4 marks)

15

A and B are points on the circumference of a circle, centre O.
AC is a tangent to the circle.
OBC is a straight line.

$OA = 5$ cm
$AC = 12$ cm



Find the length of BC.
You must show all your working.

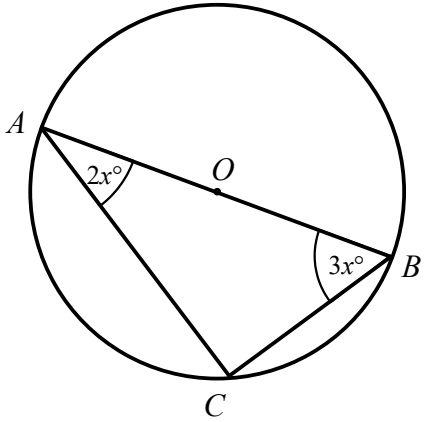
(4 marks)

16

A, B and C are points on the circumference of a circle, centre O.

Angle $CAB = 2x°$
Angle $ABC = 3x°$

Find the value of x.
You must show all your working.



(3 marks)

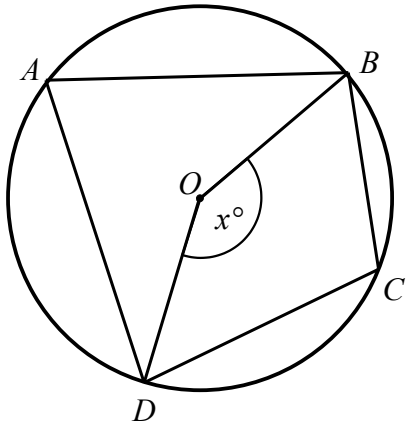
17

A, B, C and D are points on the circumference of a circle, centre O.

Angle $BOD = x°$

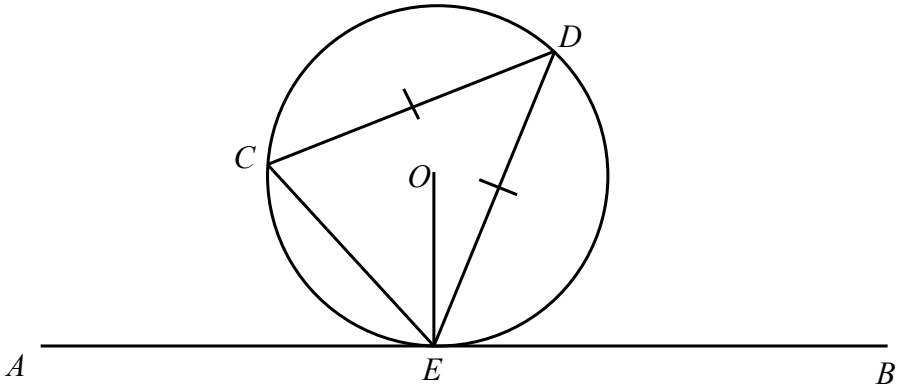
Find the size of angle BCD, in terms of x.

Give reasons for each stage of your working.



(3 marks)

18



C, D and E are points on a circle, centre O.
AEB is a tangent to the circle at E.

$CD = DE$
Angle $AEC = x°$

Find the size of angle OED, in terms of x.
Give reasons for each stage of your working.

(5 marks)